Lewis Patten

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Experience

Cyber Security Engineer II October 2025 – Present

Netcraft *London, UK*

- Independently planned, scoped, and executed penetration tests across internal web applications, Windows Active Directory, Entra ID, and AWS cloud environments, identifying 25 significant vulnerabilities including multiple critical Active Directory privilege escalation chains leading to Domain Admin compromise
- Reported findings including LAPS misconfigurations, overly privileged accounts, weak LDAP password policy, exposed credentials, and multiple XSS vulnerabilities directly to department heads, driving remediation initiatives across the AD environment and web application estate
- Designed and deployed a purpose-built security alerting platform in Go on AWS EKS using Terraform, aggregating outputs from Nessus, Semgrep, Black Duck, and Vuls into actionable alerts for engineering teams, serving as the organisation's primary vulnerability detection layer
- Led incident response for production security events including customer-reported XSS, service outages, a major third-party supply chain failure impacting CI/CD pipelines, and erroneous DNS records requiring external vendor coordination

Software Engineer August 2023 – June 2025

Garrison (now part of Everfox) *London, UK*

- Worked collaboratively to develop cutting-edge hardware security solutions for stakeholders in government, military, intelligence, and corporate sectors, enhancing protection of critical assets and sensitive information.
- Contributed to a large, rapidly evolving codebase using C++20, C, Python, and TypeScript for a highly distributed system, maintaining sophisticated modern and legacy code components
- Independently identified 6 critical vulnerabilities in the product offerings, implemented patches, and managed their testing and release to customers
- Developed a comprehensive automated test suite for a critical, bug-prone component, uncovering 5 previously undetected bugs and automating 12 time-consuming manual tests, saving multiple days during QA, and accelerating release cycles
- Made significant contributions to 6 major software releases, including as the primary developer on the first release of the software component of a next generation cross domain transfer solution
- Independently completed the platform-level transition to FIPS 140-2 cryptography, enabling FedRAMP compliance and opening up sales of our cloud offering to US federal agencies

Software Engineer June 2022 – August 2023

Darktrace *Cambridge, UK*

- Applied knowledge of cybersecurity to aid in the development of a groundbreaking endpoint detection and response (EDR) and network intrusion detection solution.
- Worked in a variety of languages including C++, Python and Swift to develop key new features in addition to maintaining and improving the existing codebase, with code being deployed to tens of thousands of endpoints.
- Made significant contribution to a complex and rapidly evolving cross platform, multi-language codebase at one of the UK's top Cyber Security firms.

Junior Software Engineer January 2022 – June 2022

4sight Imaging *Manchester, UK*

- Contributed to a modern C++ codebase working on sophisticated machine learning based image processing and analysis technologies for stakeholders in police and military.
- Implemented and trained modern statistical machine learning models on high dimensional datasets using Python libraries such as Pandas and Scikit Learn, contributing significantly to the functionality of novel product offerings.
- Worked in an agile environment following software engineering best principles to modernize and refactor a large, complex legacy codebase.

Education

University of York York, UK

*MMath (Hons) Mathematics - **1st*** *Sep. 2017 – June 2021*

Technical Skills

Security Tools: Ghidra, IDA, radare2, GDB, WinDbg, binwalk, QEMU, AFL++, Wireshark, Nessus, Semgrep, Black Duck, Nmap, Metasploit, Burp Suite, BloodHound, Impacket, pwntools

Development Tools: Linux, Git, GitLab, Jira, CMake, Docker, Protobuf, JSON, OpenAPI, Visual Studio, Xcode

Programming Languages: C++, C, Python, Go, TypeScript, JavaScript, SQL, Swift, Objective-C, R, Bash, PowerShell, x86/x86-64 Assembly, ARM Assembly, MIPS Assembly

Security Research

CVE-2026-XXXXX 2026
Unauthenticated remote buffer overflow in Tenda W150D Router, enabling RCE as root

CVE-2026-XXXXX 2026
Command injection leading to constrained shell escape in Tenda W150D Router

CVE-2026-30649 2026
Unauthenticated remote buffer overflow in Vivotek FD8136 IP cameras, enabling RCE as root.

CVE-2026-30650, CVE-2026-30652, CVE-2026-35716, CVE-2026-35717 2026
Four post-authentication buffer overflows across multiple endpoints in Vivotek FD8136 IP cameras.

CVE-2026-35718 2026
Authenticated path traversal in Vivotek FD8136 IP cameras, enabling arbitrary file read.

CVE-2025-57198 – CVE-2025-57201 2025
Four authenticated command injection vulnerabilities across multiple AVTECH IP camera web API endpoints.

CVE-2025-57202 2025
Stored XSS vulnerability in user management endpoint of AVTECH IP cameras.

Conference Talks

U-ART, U-Boot, U-Pwn: Dumping IoT Firmware the Hard Way July 2026
Hands-on workshop covering the full IoT firmware extraction pipeline

- Delivered a hands-on workshop teaching attendees to identify and connect to UART debug ports on IoT PCBs, leverage U-Boot's built-in commands to dump firmware from flash memory, and reverse engineer the extracted binaries in Ghidra

Certifications

CREST Registered Penetration Tester (CRT) – CREST Dec 2026
CREST Practitioner Security Analyst (CPSA) – CREST Apr 2026
Certified Red Team Professional for Azure (CARTP) – Altered Security Dec 2025
Certified Red Team Expert (CRTE) – Altered Security Nov 2025
Microsoft Certified: Azure Fundamentals (AZ-900) – Microsoft Oct 2025
AWS Certified Cloud Practitioner – Amazon Web Services Oct 2025
OffSec Certified Professional (OSCP) – OffSec Jun 2025
Practical Junior Penetration Tester (PJPT) – TCM Security Feb 2025
Certified Red Team Professional (CRTP) – Altered Security Oct 2023
Certified Red Team Operator (CRTO) – Zero-Point Security Feb 2023

Publications

British Journal of Cancer 2022
Sodium accumulation in breast cancer predicts malignancy and treatment response *James, Andrew D, et al.*